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MAT 524: Functions of a Real Variable II
Homework IX

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[[Problem One]]:

Problem Statement

Let $f \in L^1(\mathbb{R})$. Prove that $\hat{f}$ is a continuous function on $\mathbb{R}$.

Solution

Let $y_0 \in \mathbb{R}$. We will show that $\lim\limits_{y \to y_0} \hat{f}(y) = \hat{f}(y_0)$. By definition of the Fourier Transform, we have

$$\lim_{y \to y_0} \hat{f}(y) = \lim_{y \to y_0} \int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx. \tag{1}$$

Note that $f(x)e^{-2\pi i x y}$ is dominated by $|f(x)|$, in the sense that $|f(x)e^{-2\pi i x y}| \leq |f(x)|$. Furthermore, since $f \in L^1(\mathbb{R})$, we have $\int_{\mathbb{R}} |f(x)| dx < \infty$. Now,

$$\begin{aligned}
\lim_{y \to y_0} f(x) e^{-2\pi i x y} &= f(x) \lim_{y \to y_0} [\cos(2\pi x y) - i \sin(2\pi x y)] && (2) \\
&= f(x) [\cos(2\pi x y_0) - i \sin(2\pi x y_0)] && (3) \\
&= f(x) e^{-2\pi i x y_0}. && (4)
\end{aligned}$$

Thus, by the Lebesgue Dominated Convergence Theorem,

$$\begin{aligned}
\lim_{y \to y_0} \int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx &= \int_{\mathbb{R}} f(x) e^{-2\pi i x y_0} dx && (5) \\
&= \hat{f}(y_0). && (6)
\end{aligned}$$

Therefore, $\hat{f}$ is continuous on $\mathbb{R}$.

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[[Problem Two]]:

Problem Statement

Let $f = \chi_{[-1,1]}$ be the characteristic function of $[-1,1]$. Show that for any real number $y \neq 0$,

$$\hat{f}(y) = \frac{\sin(2\pi y)}{\pi y}.$$

Show directly that $\hat{f}$ is continuous at $y = 0$.

Solution

Let y be non-zero. Then

$$\begin{aligned}
\hat{f}(y) &= \int_{\mathbb{R}} \chi_{[-1,1]}(x) e^{-2\pi i x y} dx && (7) \\
&= \int_{-1}^{1} e^{-2\pi i x y} dx && (8) \\
&= \left[\frac{e^{-2\pi i x y}}{-2\pi i y}\right]_{x=-1}^{x=1} && (9) \\
&= \frac{e^{-2\pi i y} - e^{2\pi i y}}{-2\pi i y} && (10) \\
&= \frac{-2i \sin(2\pi y)}{-2\pi i y} && (11) \\
&= \frac{\sin(2\pi y)}{2\pi y}. && (12)
\end{aligned}$$

Next, we will show that $\hat{f}$ is continuous at $y = 0$. We have that

$$\lim_{y \to 0} \hat{f}(y) = \lim_{y \to 0} \int_{-1}^{1} e^{-2\pi i x y} dx. \tag{13}$$

Note that the function $e^{-2\pi i x y}$ is dominated by 1 in the sense that $|e^{-2\pi i x y}| \leq 1$, and $\int_{-1}^{1} 1 dx < \infty$. Furthermore,

$$\begin{aligned}
\lim_{y \to 0} e^{-2\pi i x y} &= \lim_{y \to 0} [\cos(2\pi x y) - i \sin(2\pi x y)] && (14) \\
&= \cos(0) - i \sin(0) && (15) \\
&= 1. && (16)
\end{aligned}$$

Thus, by the Lebesgue Dominated Convergence Theorem,

$$\lim_{y \to 0} \int_{-1}^{1} e^{-2\pi xy} dx = \int_{-1}^{1} e^{0} dx \tag{17}$$

$$= \hat{f}(0). \tag{18}$$

Therefore, $\hat{f}$ is continuous at 0. (It actually takes the value 2 at $y = 0$).

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$\left[\left[\text{ Problem Three }\right]\right]$:

Problem Statement

Let $f(x) = e^{-\pi x^2}$.

(a) Show that $\hat{f}(0) = 1$.

(b) Show more generally that $\hat{f}(y) = f(y)$ for all y.

Solution

(a) The Fourier transform of f is

$$\hat{f}(y) = \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi ixy} dx. \tag{19}$$

Therefore,

$$\hat{f}(0) = \int_{\mathbb{R}} e^{-\pi x^2} dx. \tag{20}$$

Now, to compute this integral, we consider what happens when we take its square:

$$\left(\int_{\mathbb{R}} e^{-\pi x^2} dx\right)^2 = \left(\int_{\mathbb{R}} e^{-\pi x^2} dx\right)\left(\int_{\mathbb{R}} e^{-\pi y^2} dy\right) \tag{21}$$

$$= \iint_{\mathbb{R}^2} e^{-\pi x^2 - \pi y^2} dxdy. \tag{22}$$

This integral can be easily computed in polar coordinates. We have

$$\iint_{\mathbb{R}^2} e^{-\pi x^2 - \pi y^2} dxdy = \int_0^{2\pi} \int_0^{\infty} e^{-\pi r^2} rdrd\theta \tag{23}$$

$$= \int_0^{2\pi} \frac{1}{2\pi} d\theta \tag{24}$$

$$= \frac{2\pi - 0}{2\pi} \tag{25}$$

$$= 1. \tag{26}$$

(b) We know from class that **(*)** $(2\pi i)M\hat{f} = \widehat{Df}$ and **(**)**$\widehat{Mf} = \frac{-1}{2\pi i} D\hat{f}$. From **(*)**, we see that

$$2\pi iy\hat{f}(y) = \int_{\mathbb{R}} -2\pi x e^{-\pi x^2} e^{-2\pi ixy} dx \tag{27}$$

$$= -2\pi \int_{\mathbb{R}} x e^{-\pi x^2} e^{-2\pi ixy} dx. \tag{28}$$

In other words, $-iy\hat{f}(y) = \widehat{Mf}$. But from **(**)**, we know that $\widehat{Mf} = \frac{-1}{2\pi i}Df$, so we have

$$-iy\hat{f}(y) = \frac{-1}{2\pi i}\hat{f}'(y). \tag{29}$$

This can be written as the separable differential equation

$$-\frac{1}{2\pi}\frac{d\hat{f}}{dy} = y\hat{f}. \tag{30}$$

The solution to this differential equation is

$$\hat{f}(y) = Ae^{-\pi y^2}. \tag{31}$$

From part **(a)**, we know that $A = 1$. Therefore, $\hat{f}(y) = f(y)$.

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